

Week 14 Class

Zachary Yildiz-Raslan

Extra Credit Section

Task 1

```
# Attach selected coefficients back to the spatial data:
boston_sf$LSTAT_local <- gwr_mod$SDF$LSTAT #first create a separate column with the predictor
# Plot

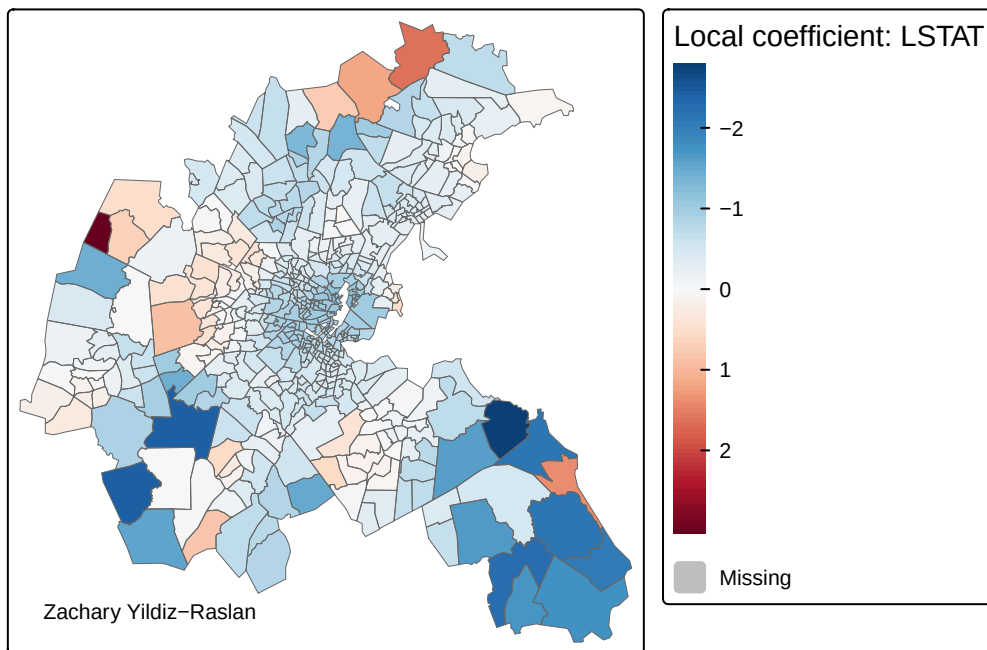
tmap_mode("plot") # static mode
```

```
i tmap modes "plot" - "view"
i toggle with `tmap::ttm()``
```

```
tm_shape(boston_sf) +
  tm_fill(
    "LSTAT_local", #put the new column made on line 246 here
    palette = "-RdBu",
    style = "cont",
    title = "Local coefficient: LSTAT" #last change the title based on the new column mapped
  ) +
  tm_credits(
    "Zachary Yildiz-Raslan",
    position = c("left","bottom")
  ) +
  tm_borders(col = "gray40", lwd = 0.5)
```

```
-- tmap v3 code detected -----
[v3->v4] `tm_fill()`: instead of `style = "cont"`, use fill.scale =
`tm_scale_continuous()`.
-----
```

i Migrate the argument(s) 'palette' (rename to 'values') to
 'tm_scale_continuous(<HERE>)'[v3->v4] `tm\_fill()`: migrate the argument(s) related to the
 visual variable `fill` namely 'title' to 'fill.legend = tm_legend(<HERE>)' Multiple palettes
 Variable(s) "fill" contains positive and negative values, so midpoint is set to 0. Set midpo.
 [cols4all] color palettes: use palettes from the R package cols4all. Run
 `cols4all::c4a\_gui()` to explore them. The old palette name "-RdBu" is named
 "rd_bu" (in long format "brewer.rd_bu")



I think this map shows enough variation in coefficient values to justify use of a local model.

```
AIC(ols, lag_mod, err_mod)
```

```
gwr_mod$results$AICc
```

```
[1] 3264.43
```

```
ols_resid <- residuals(ols)
lag_resid <- residuals(lag_mod)
err_resid <- residuals(err_mod)
gwr_resid <- gwr_mod$SDF$gwr.e

moran.test(ols_resid, nbw, zero.policy = TRUE)
```

Moran I test under randomisation

data: ols_resid
weights: nbw

Moran I statistic standard deviate = 15.527, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:

Moran I statistic	Expectation	Variance
0.4144258701	-0.0019801980	0.0007192302

```
moran.test(lag_resid, nbw, zero.policy = TRUE)
```

Moran I test under randomisation

data: lag_resid
weights: nbw

Moran I statistic standard deviate = 5.3569, p-value = 4.234e-08
alternative hypothesis: greater
sample estimates:

Moran I statistic	Expectation	Variance
0.1414596789	-0.0019801980	0.0007169982

```
moran.test(err_resid, nbw, zero.policy = TRUE)
```

Moran I test under randomisation

data: err_resid
weights: nbw

Moran I statistic standard deviate = -0.33035, p-value = 0.6294
alternative hypothesis: greater
sample estimates:

Moran I statistic	Expectation	Variance
-0.0108201584	-0.0019801980	0.0007160846

```
moran.test(gwr_resid, nbw, zero.policy = TRUE)
```

Moran I test under randomisation

```
data: gwr_resid
weights: nbw
```

Moran I statistic standard deviate = 7.4061, p-value = 6.504e-14

alternative hypothesis: greater

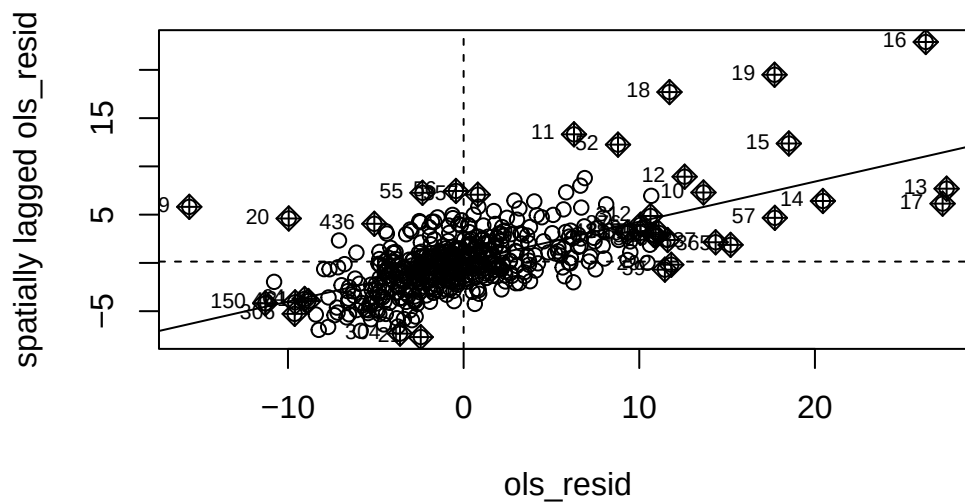
sample estimates:

Moran I statistic	Expectation	Variance
0.1968083719	-0.0019801980	0.0007204515

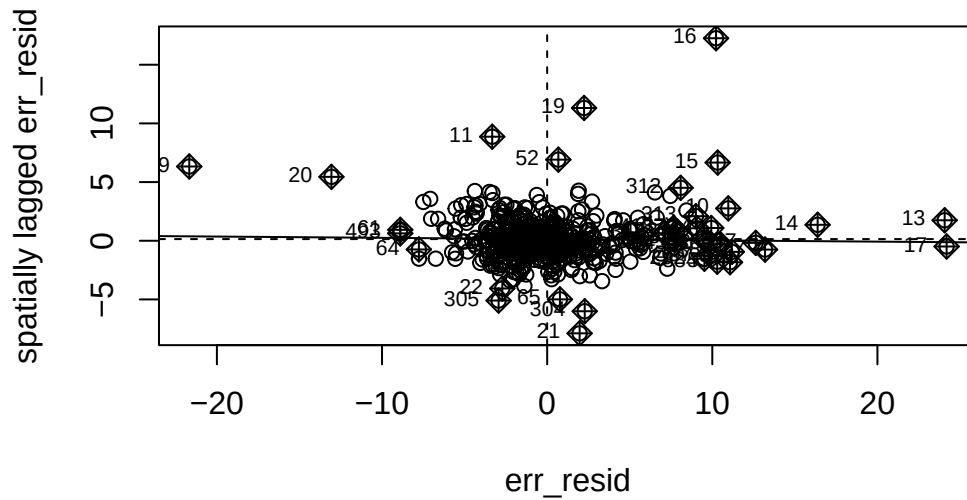
Based on the Moran's I test results, the err_resid test appears to do the best job of accounting for spatial autocorrelation due to low residual spatial autocorrelation (-0.0108201584) and non significant p-value which indicates random spatial pattern.

Task 2

```
moran.plot(ols_resid, nbw)
```



```
moran.plot(err_resid, nbw)
```



Both the OLS shows moderate positive autocorrelation and the error plot shows a very slight negative (almost 0, spatial correlation). The range of spatial lag is close for both plots.

```
par(mfrow = c(2, 2))

# OLS
ols_resid <- residuals(ols)
ols_fitted <- fitted(ols)

plot(
  ols_fitted, ols_resid,
  xlab = "Fitted values",
  ylab = "Residuals",
  main = "OLS: Residuals vs Fitted"
)
abline(h = 0, lty = 2)
lines(lowess(ols_fitted, ols_resid), col = "red", lwd = 2)

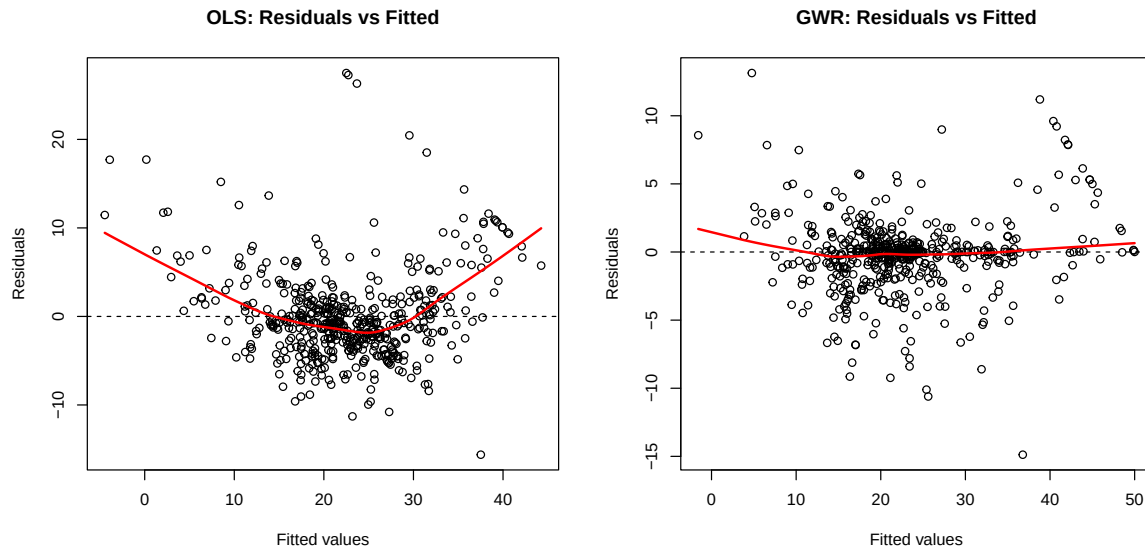
# Spatial lag

# Spatial error
```

```
# GWR
gwr_fitted <- gwr_mod$SDF$pred
gwr_resid <- gwr_mod$SDF$gwr.e

plot(
  gwr_fitted, gwr_resid,
  xlab = "Fitted values",
  ylab = "Residuals",
  main = "GWR: Residuals vs Fitted"
)
abline(h = 0, lty = 2)
lines(lowess(gwr_fitted, gwr_resid), col = "red", lwd = 2)

title(sub = "Zachary Yildiz-Raslan", outer = TRUE, line = -1)
```



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The pronounced curve in the OLS shows clear non-linearity, while the straight line in the GWR plot shows a much better model fit for the LSTAT data.

Task 3

```
par(mfrow = c(2, 2))  
  
# OLS
```

```

ols_resid <- residuals(ols)
ols_fitted <- fitted(ols)

plot(
  ols_fitted, ols_resid,
  xlab = "Fitted values",
  ylab = "Residuals",
  main = "OLS: Residuals vs Fitted"
)
abline(h = 0, lty = 2)
lines(lowess(ols_fitted, ols_resid), col = "red", lwd = 2)

# Spatial lag
lag_resid <- residuals(lag_mod)
lag_fitted <- fitted(lag_mod)

```

This method assumes the response is known - see manual page

```

plot(
  lag_fitted, lag_resid,
  xlab = "Fitted Values",
  ylab = "Residuals",
  main = "Lag: Residuals VS Fitted"
)
abline(h = 0, lty = 2)
lines(lowess(lag_fitted, lag_resid), col = "red", lwd = 2)

# Spatial error
err_resid <- residuals(err_mod)
err_fitted <- fitted(err_mod)

```

This method assumes the response is known - see manual page

```

plot(
  err_fitted, err_resid,
  xlab = "Fitted Values",
  ylab = "Residuals",
  main = "Error: Residuals VS Fitted"
)
abline(h = 0, lty = 2)

```



```

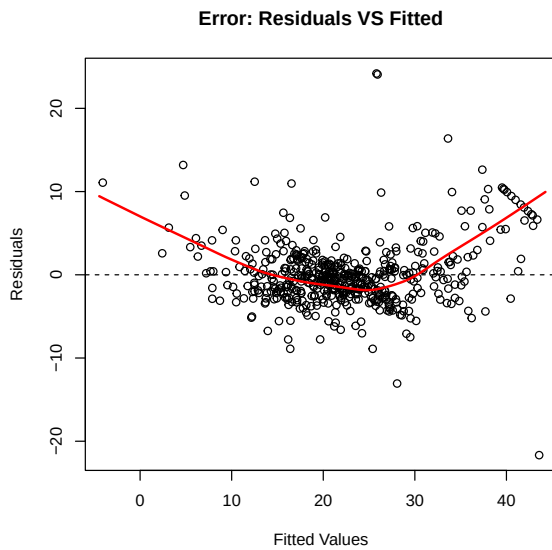
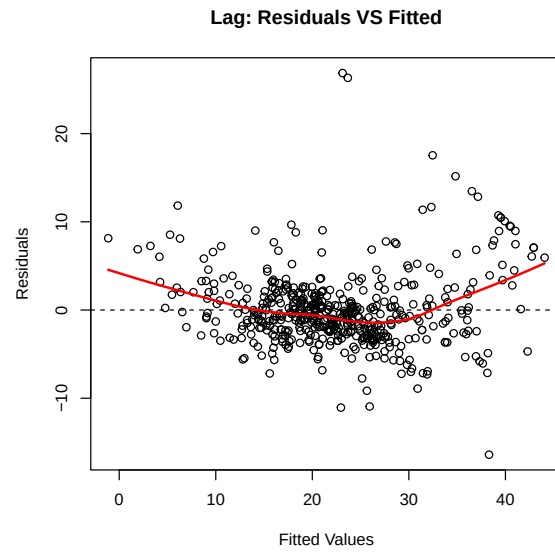
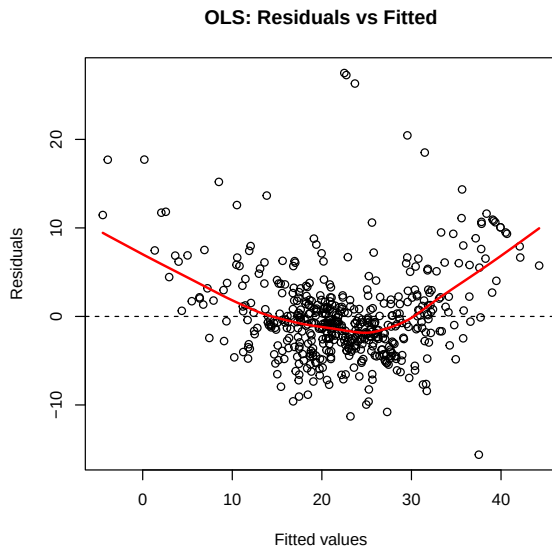
lines(lowess(ols_fitted, ols_resid), col = "red", lwd = 2)

# GWR
gwr_fitted <- gwr_mod$SDF$pred
gwr_resid <- gwr_mod$SDF$gwr.e

plot(
  gwr_fitted, gwr_resid,
  xlab = "Fitted values",
  ylab = "Residuals",
  main = "GWR: Residuals vs Fitted"
)
abline(h = 0, lty = 2)
lines(lowess(gwr_fitted, gwr_resid), col = "red", lwd = 2)

title(sub = "Zachary Yildiz-Raslan", outer = TRUE, line = -1)

```



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For my final project, I am interested in seeing how the relationship between predictor and response vary of space, so I would be interested in the GWR model, which also appears to have the lowest level of non-linearity among these four plots.